

Possibility of Acoustic Biomass Estimation by the Ordinary Echo sounder for *Cavala preta*, *Decapterus macarellus* in Cabo Verde waters

Vito Ramos^{*1} and Shunji Fujiwara^{*2}

^{*1} National Institute of Fishery Development (INDP), Mindelo, S. Vicente,
the Republic of Cabo Verde

^{*2} Overseas Fishery Cooperation Foundation(OFCF), Tokyo, Japan

Abstract

In order to examine possibility of biomass estimation for *Cavala preta*, *Decapterus macarellus*, using ordinary echo sounder, echograms were confirmed. Average size of fish school was estimated using R/V Sinagoga purse seine operations. Estimated school size was approximately 1.7 tons. Based on these information, biomass estimation could be possible, when number of fish school and cruising distance are obtained through transect cruising.

Introduction

Acoustic method of biomass estimation has been a routine work for resource survey. At present, so called “Quantitative Echo Sounder” is the most commonly used. The acoustic theory is simply that received echo strength is proportional to the density of fish (Foote, 1983). If a single fish echo strength, which is Target Strength, is known, fish density is obtained from observed echo strength. In Cabo Verde, two acoustic biomass estimations by the quantitative echo sounder were carried out (Stromme *et al.*, 1982; IPIMAR and INDP, 1994)

The quantitative echo sounder is a quite bulky and expensive, moreover handling of the instrument is complicated, although it is called considered to be more scientific approach. In some condition, for example, in shallow water area, it is sometimes difficult to operate in that area. If it is necessary to estimate density of fish by the acoustic method, other alternative acoustic method should be introduced. It can be a counting method, using the ordinary echo sounder, however this can compromise some precision. This is solely to count fish school or a single fish. If volume of water to be scanned by the echo sounder is known, density of fish can be estimated.

There are two things to be clarified to estimate biomass, using the ordinary echo sounder: one is echo identification and the other is size of school.

For both the quantitative and ordinary echo sounder, species identifications of the echograms are necessary through direct observation by the diving or indirect by the fishing.

From the quantitative echo sounder, numerical output is linearly related to a size of fish school(Foote *et al.*; 1983). However, none of information can not be withdrawn from the ordinary echo sounder, thus size of fish school should be obtained from other source of information.

This survey is focused on species identification of echogram and estimation of size of fish school in order to examine the possible biomass estimation for *Cavala preta*, *Decapterus macarellus*, in Cabo Verdian waters, using the ordinary fish finder.

MATERIALS AND METHODS

It is necessary to extract some characteristics of the echogram from *Cavala preta* in order to identify echograms of *Cavala preta* from the other species. When echogram seemed to be *Cavala preta*, fishing operations were made for species identification corresponding to the echogram. Throughout series of observations, echograms of *Cavala preta* would be confirmed. At the same time, echograms were recorded by the camera.

A density of fish school was obtained, using records of fishing operation survey by R/V Sinagoga from November 2002 to October 2003. After operation, sub-samples of Cavala preta were taken, and length and weight were recorded. Using this information, an average weight or number of fish school was calculated.

Results

As results of series of observations of echogram, using the ordinary echo-sounder of R/V Sinagoga, the echogram from Cavala preta will be characterized. As showing in Figure 1, typical echogram of Cavala preta is oval shape.

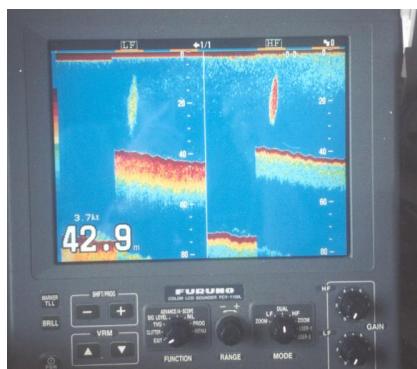


Figure 1. Typical echogram of Cavala preta around S. Vicente, using FCV1100L.

Table 1 showed results of purse seine operations from November 2002 to October 2003, using R/V Sinagoga.

Table 1. Average number and weight of captured Cavala preta by the purse seine of R/V Sinagoga from November 2002 to October 2003.

Year	Month	Cavala Landing (kg)	Effective Effort	Catch per Effort (kg)	Avg.wt(g)	No. fish
2002	November	4074	3	1,358	107	12703
	December	5,737	6	956	136	7016
2003	January	9978	9	1,109	189	5870
	February	13736	5	2,747	241	11380
	March	9453	4	2,363	246	9619
	April	7662	5	1,532	231	6641
	May	17,708	8	2,214	191	11601
	June	289.5	1	290	307	944
	July	3028.4	1	3,028	289	10477
	August	9,560	7	1,366	280	4883
	September	2,800	1	2,800	171	16388
	October	3823.6	4	955.9	330	2900
				1,727		8368

Average Cavala preta catch per one operation was 1.7 tons. Based on length-weight relationship of Cavala preta, average weight of individuals was calculated. As result of this, number of Cavala preta was approximately 8,000 fish per operation.

Half angle of transducer of 50kHz is 12°. In case of searching depth 100m, average searching width is 0.0213 km. When survey cruise is made, cruising distance and number of Cavala preta will be obtained.

Biomass estimation can be in the following calculation:

Density of fish school = (number of echogram) / (searching width) x (length of cruising)

Biomass in weight = (Density of fish school) x (average weight of fish school) x (target area) .

This report showed characteristics of echograms and average size for Cavala preta. Based on these information, biomass can be estimated after cruising.

Survey will be necessary monthly or yearly to examine trend of biomass. Previously, two acoustic surveys were carried out in 1981 to 1982, and 1997. Results of R/V Sinagoga can be compared with these previous results to see condition of Cavala preta.

Literatures

Foote, K. G. 1983. Linearity of fisheries acoustics, with addition theorems. J. Acoust. Soc. Am., 73:1932-40

IPIMAR and INDP. 1997. Campanha de oceanographia e avaliacao de pelagicos na ZEE de Cabo Verde Junho/Julho 1997 NI-Capricornio. Relat. Cient. Tec. Inst. Invest. Pescas e Mar, 69p.

Stromme, T., S. Sundby, and G. Saetersdal. 1982. A survey of the fish resources in the coastal waters of the Republic of Cape Verde. Nov. 1981. In: 1984: Report on the R/V DR. Fridtjof Nansen fish resource surveys off West Africa: Morocco to Ghana and Cape Verde. May 1981-March 1982. CEFAF/ECAF Series 84/29 Rome.